BEAM DESIGN CALCULATION

Project Name	:	G+2 RB BEAM DESIGN @FLOOR 2
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-8.077 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	8.077 m

Sr.No.	Symbol		Definitions
1	α	=	Angle formed with horizontal by diagonal reinforcement
2	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
3	εm	=	Average steel strain at level considered
4	Acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
5	Adr	=	Area of diagonal reinforcement in coupler beam
6	AstCalc	=	Area of Tension reinforcement required in sqmm
7	Ast comp	=	Ast at compression side
8	AstCrack	=	Area of Tension reinforcement for Crack Width required in sqmm
9	AstPrv	=	Area of reinforcement provided at given section in sqmm
10	Asv	=	Area of shear reinforcement required per meter length in sqmm
11	Asv Reqd	=	Min area shear reinforcement as per clause 41.4.3
12	Asv Total Prv	=	Total area of shear reinforcement provided, including that for torsion
13	Asv Torsion	=	Area of Stirrups over the corner bars to resist shear and torsion as per clause 41.4.3
14	Asv Torsion Prv	=	Area of shear reinforcement resisting torsion (outer legs only)
15	b	=	Width of the Beam in mm
16	bw	=	Width of Web in mm
17	Ceff	=	Effective Cover of Beam in mm
18	Cmin	=	Clear cover to top and bottom reinforcement in mm
19	Cside	=	Clear cover to side reinforcement in mm
20	d	=	Effective depth of Beam in mm
21	d'	=	Clear cover to reinforcement at compression side.
22	D	=	Depth of Beam in mm
23	D'	=	Distance (c/c) between top-most and bottom-most layer
24	Df	=	Thickness of Flange in mm
25	Ec	=	Modulus of elasticity of concrete
26	Es	=	Modulus of elasticity of steel
27	FcPerm	=	Permissible Stress in Concrete required in N/sqmm

28 Fst = Stress in steel
29 FstPerm = Permissible Stress in Steel required in N/sqmm
30 Legs = Number of legs of the shear reinforcement
31 lo = Effective Length of Beam in mm
32 Mh = Hogging Moment of Resistance in kNm
33 Ms = Sagging Moment of Resistance in kNm
34 M_{Tu} = Converted Bending Moment due to Torsion at a section in kNm
35 Mu = Factored Bending Moment in Limit State at a section in kNm
36 Mud = Total Design Moment = Mu + M_{Tu} in kNm
37 MuLim = Limiting Moment of Resistence of Singly Reinforced Section
38 NA angle (φ) = It is a neutral axis angle corresponding to load angle to find out MCap
39 Pcclc = Calculated percentage compression steel
40 Ptclc = Calculated percentage tension steel
41 Ptmin = Minimum percentage steel as per code
42 PtPrv = provided percentage steel
43 R = Moment Ratio Mud/Bd^2
44 Reinf = Rebar arrangement provided
45 sp = Spacing Between bars at outer most layer
46 Spc1 to Spc6 = Max Stirrup Criteria in mm
47 Stirrup Dia = Diameter of shear stirrups in mm
48 SvCalc = Stirrup spacing calculated as per Asv in mm
49 SvPrv = Stirrup spacing provided in m
50 Tc = Permissible Shear stress in concrete at a given section in N/sqmm Calculated as per Tensile reinforcement
51 Tce = Earthquake inducted shear stress in N/sqmm
52 Tcep = Permissible value of shear stress for checking of coupling action in N/sqmm
53 Tu = Factored Torsional Moment in Limit State at a section in kNm
54 Tv = Shear stress due to Vud in N/sqmm
55 V^{D+L} = Shear force 1.2x(Dead Load + Live Load) in kN
56 Vc = Shear capacity of concrete section due to Tc in kN
57 Vu = Factored Shear Force in Limit State at a section in kN
58 Vu-A1 = V^{D+L}Left - 1.4 x (MsLeft + MhRight)/L in kN
59 Vu-A2 = V^{D+L}Left + 1.4 x (MhLeft + MsRight)/L in kN
60 Vu-B1 = V^{D+L}Right + 1.4 x (MsLeft + MhRight)/L in kN
61 Vu-B2 = V^{D+L}Right - 1.4 x (MhLeft + MsRight)/L in kN
62 Vud = Design Shear Force (Max of Vus and Vu-sway) in kN
63 Vue = Earthquake inducted shear in kN
64 Vu-sway = Max of (Vu-A1 and Vu-A2) & (Vu-B1 and Vu-B2) in kN
65 Vus = Vud - Vc Design Shear Force for reinforcement, in kN
66 Vut = Total Shear Force = Vu + V_{Tu} in kN

67 V_{Tu} = Converted Shear Force due to Torsion at a section in kN

68 Wcr = Surface Crack Width

69 WcrPerm = Permissible Crack Width required in mm

70 X1 = Smaller c/c dimension of outermost Stirrups

71 Y1 = Larger c/c dimension of outermost Stirrups

All 6mm diameter bars are assumed 250 grade with fy = 250 N/sqmm

All Forces are in 'kN', 'kNm', Stress in 'N/sqmm' & Dimension are in 'mm'.

Code References

IS 456

Sr.No.	Item	Clause / Table
1.	Ptmax	26.5.1.1
2.	Ptmin	26.5.1.1
3.	M_{Tu}	41.4.2
4.	V_{Tu}	41.3.1
5.	Ptclc	38
6.	Tc	40.2.1
7.	Tcmax	40.2.3
8.	Asv	40.4
9.	Min Shear Reinf	26.5.1.6
10.	Max Stirrup Spacing	26.5.1.5
11.	Shear Reinf - Torsion	26.5.1.7
12.	Side Face Reinforcement	26.5.1.3
13.	crack width calculation	Annex-F
14.	cracking	Clause 35.3.2

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Sr.No.	Item	Clause / Table
1.	Ptmax	6.2.2
2.	Ptmin	6.2.1
3.	Astmin	6.2.3 & 6.2.4
4.	Svclc	6.3.5
5.	Shear Capacity	6.3.4
6.	Coupling Beams	10.5

Group : **G1**

Beam No : **B1**

Analysis Reference (Member) : 5 (8.077 m)

Beam Length : 3924 mm

Breadth (B) : 230 mm

Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	56.01	46.76	23.39	61.23	23.5	81.05
Tu (kNm)	0.42	0.53	0	0	0	0.53
M_{Tu} (kNm)	0.71	0.89	0	0	0	0.89
Mud (kNm)	56.72	47.65	23.39	61.23	23.5	81.94
MuLim (kNm)	102	102	102	102	102	102
R	1.801	1.513	0.743	1.945	0.746	2.602
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.414	0.342	0.2	0.451	0.2	0.632
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.473	0.709
AstCalc (sqmm)	352.66	291.15	170.2	384.17	170.2	538.07
AstPrv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
				Area of reinforcement required for

D = AstCrack = Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	0.709	0.473	0.709
Vu (kN)	53.23	46.38	69.94
Tu (kNm)	0	0.53	0.53
V_{Tu} (kN)	0	3.67	3.67
Vut (kN)	53.23	50.05	73.61
Tv (N/sqmm)	0.63	0.59	0.86
Tc (N/sqmm)	0.56	0.48	0.56
Vc (kN)	47.85	40.65	47.85
Vus=Vut-Vc (kN)	5.38	9.4	25.76
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	171.952	245.67
Asv Torsion Prv (sqmm)	-	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	140
SvPrv (mm)	275	275	140
Asv Total Prv (sqmm)	365.6	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G1**	
Beam No	:	**B2**	
Analysis Reference (Member)	:	6 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	14	19	14	17	12
Critical L/C - RCDC	13	10	15	10	13	8
Mu (kNm)	31.4	31.39	39.57	64.19	30.94	83.48
Tu (kNm)	0	0.49	0.38	0.49	0	0
M_{Tu} (kNm)	0	0.83	0.64	0.83	0	0
Mud (kNm)	31.4	32.21	40.21	65.02	30.94	83.48
MuLim (kNm)	102	102	102	102	102	102
R	0.997	1.023	1.277	2.065	0.982	2.651
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.219	0.225	0.285	0.483	0.216	0.647
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.473	0.709
AstCalc (sqmm)	186.35	191.48	242.41	411.14	183.48	550.19
AstPrv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
Maximum of % reinforcement required for

B'	=	Ptclc	=	Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	0.709	0.473	0.709
Vu (kN)	58.98	38.97	62.52
Tu (kNm)	0.49	0	0
V_{Tu} (kN)	3.41	0	0
Vut (kN)	62.39	38.97	62.52
Tv (N/sqmm)	0.73	0.46	0.73
Tc (N/sqmm)	0.56	0.48	0.56
Vc (kN)	47.85	40.65	47.85
Vus=Vut-Vc (kN)	14.54	0	14.67
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	209.46	-	-
Asv Torsion Prv (sqmm)	718.14	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm

Web Depth	= 430 <= 750 mm
Side Face Reinforcement not required.	

Group	:	**G1**	
Beam No	:	**B3**	
Analysis Reference (Member)	:	141 (8.077 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	15	15	15	15
Critical L/C - RCDC	11	11	11	11	11	11
Mu (kNm)	-	-	-	34.87	25.3	8.81
Tu (kNm)	12.12	12.12	12.12	12.12	12.12	12.12
M_{Tu} (kNm)	20.46	20.46	20.46	20.46	20.46	20.46
Mud (kNm)	20.46	20.46	20.46	55.33	45.76	29.27
MuLim (kNm)	102	102	102	102	102	102
R	0.65	0.65	0.65	1.757	1.453	0.93
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.403	0.327	0.203
Pcclc (%) (C)	0.14	0.14	0.14	0	0	0.079
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	343.11	278.65	173.12
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	15	15	15
Critical L/C - RCDC	11	11	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	29.19	25.76	22.33
Tu (kNm)	12.12	12.12	12.12
V_{Tu} (kN)	84.32	84.32	84.32
Vut (kN)	113.51	110.08	106.65
Tv (N/sqmm)	1.33	1.29	1.25
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	65.66	62.23	58.8
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	707.171	696.433	685.694
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	491.56	465.87	440.18
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic
Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion
(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B4**	
Analysis Reference (Member)	:	25 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	19	19	19	12
Critical L/C - RCDC	8	8	15	15	15	8
Mu (kNm)	68.35	36.49	33.62	58.99	28.04	73.43
Tu (kNm)	0.21	0.21	0.2	0.2	0.2	0.21
M_{Tu} (kNm)	0.36	0.36	0.34	0.34	0.34	0.36
Mud (kNm)	68.71	36.85	33.96	59.33	28.38	73.79
MuLim (kNm)	102	102	102	102	102	102
R	2.182	1.17	1.078	1.884	0.901	2.344
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.515	0.259	0.238	0.436	0.2	0.559

Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.473	0.709
AstCalc (sqmm)	437.89	220.8	202.45	370.85	170.2	475.57
AstPrv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	9	12	12
Critical L/C - RCDC	5	8	8
PtPrv (%)	0.709	0.473	0.709
Vu (kN)	38.49	42.03	53.8
Tu (kNm)	0.19	0.21	0.21
V_{Tu} (kN)	1.35	1.48	1.48
Vut (kN)	39.84	43.51	55.29
Tv (N/sqmm)	0.47	0.51	0.65
Tc (N/sqmm)	0.56	0.48	0.56
Vc (kN)	47.85	40.65	47.85
Vus=Vut-Vc (kN)	0	2.86	7.44
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	130.31	142.356	179.216
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B5**	
Analysis Reference (Member)	:	26 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	42.43	28.22	44.64	55.51	42.52	87.08
Tu (kNm)	0.19	0.16	0.16	0.15	0.2	0.2

M_{Tu} (kNm)	0.32	0.27	0.27	0.26	0.33	0.33
Mud (kNm)	42.75	28.49	44.91	55.76	42.86	87.42
MuLim (kNm)	102	102	102	102	102	102
R	1.358	0.905	1.426	1.771	1.361	2.776
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.304	0.2	0.321	0.407	0.305	0.683
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.473	0.709
AstCalc (sqmm)	258.89	170.2	273.05	346.1	259.59	581.65
AstPrv (sqmm)	603.18	603.18	603.18	603.18	402.12	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	2-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	7
Critical L/C - RCDC	10	8	3
PtPrv (%)	0.709	0.473	0.709
Vu (kN)	40.49	35.12	51.66
Tu (kNm)	0.15	0.2	0.19
V_{Tu} (kN)	1.06	1.38	1.31
Vut (kN)	41.55	36.5	52.96
Tv (N/sqmm)	0.49	0.43	0.62
Tc (N/sqmm)	0.56	0.48	0.56
Vc (kN)	47.85	40.65	47.85
Vus=Vut-Vc (kN)	0	0	5.11
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	134.465	119.987	171.213
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81

SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**	
Beam No	:	**B6**	
Analysis Reference (Member)	:	146 (8.077 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	5	8	13
Critical L/C - RCDC	9	9	9	1	4	9
Mu (kNm)	-	-	2.82	74.34	43.79	11.22
Tu (kNm)	11.07	11.07	11.07	2.47	9.45	11.07
M_{Tu} (kNm)	18.68	18.68	18.68	4.16	15.95	18.68
Mud (kNm)	18.68	18.68	21.51	78.5	59.74	29.9
MuLim (kNm)	102	102	102	102	102	102
R	0.593	0.593	0.683	2.493	1.897	0.949
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.601	0.439	0.208
Pcclc (%) (C)	0.128	0.128	0.108	0	0	0.05
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	511.3	373.72	177
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	58.66	57.84	55.25
Tu (kNm)	2.47	2.47	2.47
V_{Tu} (kN)	17.15	17.15	17.15
Vut (kN)	75.82	74.99	72.41
Tv (N/sqmm)	0.89	0.88	0.85
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85

Vus=Vut-Vc (kN)	27.97	27.14	24.56
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	308.882	306.292	298.211
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G3**	
Beam No	:	**B7**	
Analysis Reference (Member)	:	45 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm

(Cmin)
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	12	19	19	19	12
Critical L/C - RCDC	8	8	15	15	15	8
Mu (kNm)	67.63	35.44	38.39	60.62	28.07	75.14
Tu (kNm)	0	0	0.33	0.33	0.33	0
M_{Tu} (kNm)	0	0	0.55	0.55	0.55	0
Mud (kNm)	67.63	35.44	38.94	61.17	28.62	75.14
MuLim (kNm)	102	102	102	102	102	102
R	2.148	1.126	1.237	1.943	0.909	2.386
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.505	0.249	0.275	0.451	0.2	0.571
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.738	0.738	0.738
AstCalc (sqmm)	430.04	211.84	234.2	383.8	170.2	485.72
AstPrv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement Provided	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	9	7	7
Critical L/C - RCDC	5	3	3
PtPrv (%)	0.738	0.738	0.738

Vu (kN)	41.92	42.6	56.53
Tu (kNm)	0.38	0	0
V_{Tu} (kN)	2.62	0	0
Vut (kN)	44.54	42.59	56.53
Tv (N/sqmm)	0.52	0.5	0.66
Tc (N/sqmm)	0.57	0.57	0.57
Vc (kN)	48.62	48.62	48.62
Vus=Vut-Vc (kN)	0	0	7.91
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	152.548	-	-
Asv Torsion Prv (sqmm)	365.6	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G3**	
Beam No	:	**B8**	
Analysis Reference (Member)	:	46 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm

Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	48	28.63	46.47	58.75	43.81	90.84
Tu (kNm)	0.4	0	0	0.21	0.56	0.56
M_{Tu} (kNm)	0.68	0	0	0.35	0.94	0.94
Mud (kNm)	48.68	28.63	46.47	59.1	44.75	91.78
MuLim (kNm)	102	102	102	102	102	102
R	1.546	0.909	1.476	1.877	1.421	2.915
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.35	0.2	0.333	0.434	0.32	0.726
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.738	0.738	0.738
AstCalc (sqmm)	298.02	170.2	283.32	369.23	271.97	617.47
AstPrv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement Provided	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	7

Critical L/C - RCDC	10	8	3
PtPrv (%)	0.738	0.738	0.738
Vu (kN)	41.66	37.65	53.15
Tu (kNm)	0.21	0.56	0.57
V_{Tu} (kN)	1.45	3.88	3.99
Vut (kN)	43.11	41.52	57.13
Tv (N/sqmm)	0.51	0.49	0.67
Tc (N/sqmm)	0.57	0.57	0.57
Vc (kN)	48.62	48.62	48.62
Vus=Vut-Vc (kN)	0	0	8.51
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	142.888	148.57	198.464
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G3**
Beam No : **B9**
Analysis Reference (Member) : 151 (8.077 m)
Beam Length : 1372 mm

Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Cantilever Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	13	13	13	5	8	13
Critical L/C - RCDC	9	9	9	1	4	9
Mu (kNm)	-	-	-	64.66	36.6	11.12
Tu (kNm)	12.68	12.68	12.68	4.39	11.19	12.68
M_{Tu} (kNm)	21.41	21.41	21.41	7.41	18.89	21.41
Mud (kNm)	21.41	21.41	21.41	72.08	55.49	32.53
MuLim (kNm)	102	102	102	102	102	102
R	0.68	0.68	0.68	2.289	1.762	1.033
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.544	0.404	0.227
Pcclc (%) (C)	0.147	0.147	0.147	0	0	0.069
PtPrv (%)	0.709	0.709	0.709	0.738	0.738	0.738
AstCalc (sqmm)	170.2	170.2	170.2	462.73	344.2	193.47
AstPrv (sqmm)	603.18	603.18	603.18	628.32	628.32	628.32
Reinforcement Provided	3-T16	3-T16	3-T16	2-T20	2-T20	2-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required

D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.738	0.738	0.738
Vu (kN)	49.64	48.82	46.23
Tu (kNm)	4.39	4.39	4.39
V_{Tu} (kN)	30.55	30.55	30.55
Vut (kN)	80.19	79.36	76.78
Tv (N/sqmm)	0.94	0.93	0.9
Tc (N/sqmm)	0.57	0.57	0.57
Vc (kN)	48.62	48.62	48.62
Vus=Vut-Vc (kN)	31.57	30.74	28.16
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	388.796	386.176	378.003
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B10**	
Analysis Reference (Member)	:	65 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	12	19	14	19	12
Critical L/C - RCDC	13	8	15	10	15	8
Mu (kNm)	64.04	50.49	35.3	70.56	27.28	89.24
Tu (kNm)	0	0.15	0.42	0.56	0.42	0.15
M_{Tu} (kNm)	0	0.26	0.71	0.95	0.71	0.26
Mud (kNm)	64.04	50.75	36.01	71.52	27.99	89.49
MuLim (kNm)	102	102	102	102	102	102
R	2.034	1.612	1.144	2.271	0.889	2.842
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.475	0.367	0.253	0.539	0.2	0.703
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	404.17	311.91	215.46	458.59	170.2	598.55
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	58.88	50.43	73.99
Tu (kNm)	0.56	0.15	0.15
V_{Tu} (kN)	3.93	1.05	1.05
Vut (kN)	62.81	51.49	75.04
Tv (N/sqmm)	0.74	0.6	0.88
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	6.12	0	18.35
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	216.164	167.629	242.19
Asv Torsion Prv (sqmm)	718.14	365.6	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	140
SvPrv (mm)	140	275	140
Asv Total Prv (sqmm)	718.14	365.6	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm

Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B11**	
Analysis Reference (Member)	:	66 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	17	19	19	14	12	12
Critical L/C - RCDC	13	15	15	10	8	8
Mu (kNm)	43.37	31.93	43.88	73.93	39.02	98.6
Tu (kNm)	0.13	0	0	0.14	0.2	0.2
M_{Tu} (kNm)	0.22	0	0	0.24	0.34	0.34
Mud (kNm)	43.6	31.93	43.88	74.17	39.36	98.93
MuLim (kNm)	102	102	102	102	102	102
R	1.385	1.014	1.394	2.355	1.25	3.142
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.311	0.223	0.313	0.562	0.278	0.797
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	264.4	189.7	266.25	478.37	236.89	678.35
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	61.87	46.02	69.57
Tu (kNm)	0.14	0.2	0.2
V_{Tu} (kN)	0.97	1.39	1.39
Vut (kN)	62.85	47.41	70.96
Tv (N/sqmm)	0.74	0.56	0.83
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	6.16	0	14.27
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	203.234	156.194	230.754
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G4**
Beam No : **B12**
Analysis Reference (Member) : 156 (8.077 m)
Beam Length : 1372 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Cantilever Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	15	15	13	8	8	15
Critical L/C - RCDC	11	11	9	4	4	11
Mu (kNm)	-	-	6.66	59.1	41.87	7.82
Tu (kNm)	6.52	6.52	4.94	3.81	3.81	6.52
M_{Tu} (kNm)	11.01	11.01	8.34	6.43	6.43	11.01
Mud (kNm)	11.01	11.01	15	65.53	48.3	18.83
MuLim (kNm)	102	102	102	102	102	102
R	0.35	0.35	0.476	2.081	1.534	0.598
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.487	0.347	0.2

Pcclc (%) (C)	0.074	0.074	0.011	0	0	0.021
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	414.86	295.52	170.2
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	8	8
Critical L/C - RCDC	1	4	4
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	53.67	48.43	47.23
Tu (kNm)	0.96	3.81	3.81
V_{Tu} (kN)	6.72	26.52	26.52
Vut (kN)	60.39	74.95	73.75
Tv (N/sqmm)	0.71	0.88	0.87
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	3.7	18.26	17.06
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	220.813	354.426	350.612
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B13**	
Analysis Reference (Member)	:	134 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	5	5	9	5	5
Critical L/C - RCDC	3	1	1	5	1	1
Mu (kNm)	14.24	17.23	9.23	6.64	-	9.67
Tu (kNm)	2.38	2.87	2.87	2.22	2.87	2.87

M_{Tu} (kNm)	4.02	4.85	4.85	3.74	4.85	4.85
Mud (kNm)	18.26	22.07	14.08	10.37	4.85	14.52
MuLim (kNm)	102	102	102	102	102	102
R	0.58	0.701	0.447	0.33	0.154	0.461
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.032	0
PtPrv (%)	0.709	0.709	0.709	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement Provided	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	7	5
Critical L/C - RCDC	1	3	1
PtPrv (%)	0.266	0.709	0.266
Vu (kN)	19.49	6.81	23.68
Tu (kNm)	2.87	2.38	2.87
V_{Tu} (kN)	19.98	16.56	19.98
Vut (kN)	39.48	23.37	43.66
Tv (N/sqmm)	0.46	0.27	0.51
Tc (N/sqmm)	0.37	0.56	0.37
Vc (kN)	31.88	47.85	31.88
Vus=Vut-Vc (kN)	7.6	0	11.78
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	206.938	142.236	220.022
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81

SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B14**	
Analysis Reference (Member)	:	135 (8.077 m)	
Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	9	9	14	9	12	12
Critical L/C - RCDC	5	5	10	5	8	8

Mu (kNm)	4.97	10.68	7.39	9.5	9.81	25.91
Tu (kNm)	1.9	1.9	1.78	1.9	0.28	0.28
M_{Tu} (kNm)	3.21	3.21	3	3.21	0.46	0.46
Mud (kNm)	8.18	13.89	10.39	12.71	10.28	26.37
MuLim (kNm)	102	102	102	102	102	102
R	0.26	0.441	0.33	0.404	0.326	0.838
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement Provided	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	9	5
Critical L/C - RCDC	1	5	1
PtPrv (%)	0.266	0.709	0.266
Vu (kN)	19.16	7.44	24.01
Tu (kNm)	1.63	1.9	1.63
V_{Tu} (kN)	11.31	13.22	11.31
Vut (kN)	30.47	20.66	35.32
Tv (N/sqmm)	0.36	0.24	0.41
Tc (N/sqmm)	0.37	0.56	0.37
Vc (kN)	31.88	47.85	31.88
Vus=Vut-Vc (kN)	0	0	3.44
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	142.56	119.846	157.72

Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G5**	
Beam No	:	**B15**	
Analysis Reference (Member)	:	159 (8.077 m)	
Beam Length	:	1372	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**

Critical L/C - Analysis	7	7	7	7	7	7
Critical L/C - RCDC	3	3	3	3	3	3
Mu (kNm)	-	-	-	18.5	14.45	9.17
Tu (kNm)	9.52	9.52	9.52	9.52	9.52	9.52
M_{Tu} (kNm)	16.07	16.07	16.07	16.07	16.07	16.07
Mud (kNm)	16.07	16.07	16.07	34.57	30.51	25.23
MuLim (kNm)	102	102	102	102	102	102
R	0.51	0.51	0.51	1.098	0.969	0.801
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.242	0.213	0.2
Pcclc (%) (C)	0.109	0.109	0.109	0	0.011	0.046
PtPrv (%)	0.709	0.709	0.709	0.266	0.266	0.266
AstCalc (sqmm)	170.2	170.2	170.2	206.3	180.84	170.2
AstPrv (sqmm)	603.18	603.18	603.18	226.2	226.2	226.2
Reinforcement Provided	3-T16	3-T16	3-T16	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	12	12	12
Critical L/C - RCDC	8	8	8
PtPrv (%)	0.266	0.266	0.266
Vu (kN)	13.89	10.45	7.02
Tu (kNm)	8.92	8.92	8.92
V_{Tu} (kN)	62.07	62.07	62.07
Vut (kN)	75.95	72.52	69.09
Tv (N/sqmm)	0.89	0.85	0.81
Tc (N/sqmm)	0.37	0.37	0.37
Vc (kN)	31.88	31.88	31.88
Vus=Vut-Vc (kN)	44.07	40.64	37.21
Legs	2	2	2

Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	496.742	486.003	475.265
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	329.93	304.24	278.56
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B16**	
Analysis Reference (Member)	:	126 (8.077 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm

Es : 2x10^5 N/sqmm

Ptnominal (%) (Bn) : 0.2 %

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	14	14	9	9	5
Critical L/C - RCDC	10	10	10	5	5	1
Mu (kNm)	-	-	-	12.5	40.21	67.01
Tu (kNm)	7.4	7.4	7.4	6.63	6.63	1.47
M_{Tu} (kNm)	12.48	12.48	12.48	11.2	11.2	2.49
Mud (kNm)	12.48	12.48	12.48	23.7	51.41	69.49
MuLim (kNm)	102	102	102	102	102	102
R	0.396	0.396	0.396	0.753	1.633	2.207
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.372	0.521
Pcclc (%) (C)	0.084	0.084	0.084	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	170.2	316.36	443.67
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)

Bn = Ptnominal = Nominal % area of reinforcement

C = Pcclc = % area of compression reinforcement required

D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	9	9	9
Critical L/C - RCDC	5	5	5
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	22.59	29.36	29.36
Tu (kNm)	6.64	6.64	6.64
	46.16	46.16	46.16

V_{Tu} (kN)			
Vut (kN)	68.74	75.51	75.51
Tv (N/sqmm)	0.81	0.89	0.89
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	20.89	27.66	27.66
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	407.772	428.954	428.954
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G6**
Beam No : **B17**
Analysis Reference (Member) : 111 (8.077 m)
Beam Length : 4496 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456

Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	24.33	27.64	28.27	81.25	22.21	65.12
Tu (kNm)	0.46	0.37	0.17	0.37	0.37	0.66
M_{Tu} (kNm)	0.77	0.63	0.28	0.63	0.63	1.12
Mud (kNm)	25.1	28.27	28.55	81.88	22.84	66.24
MuLim (kNm)	102	102	102	102	102	102
R	0.797	0.898	0.907	2.6	0.725	2.104
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.632	0.2	0.494
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.2	170.2	537.52	170.2	419.99
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	62.66	35.67	56.61
Tu (kNm)	0.37	0.37	0.66
V_{Tu} (kN)	2.59	2.59	4.61
Vut (kN)	65.25	38.27	61.22
Tv (N/sqmm)	0.77	0.45	0.72
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	17.4	0	13.37
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	215.04	130.585	210.86
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G6**
Beam No : **B18**

Analysis Reference (Member)	:	99 (8.077 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	13	13	13	15
Critical L/C - RCDC	16	9	9	9	9	11
Mu (kNm)	17.45	28.58	25.1	85.91	18.9	66.69
Tu (kNm)	0	0.17	0.17	0.17	0.17	0
M_{Tu} (kNm)	0	0.29	0.29	0.29	0.29	0
Mud (kNm)	17.45	28.87	25.39	86.2	19.19	66.69
MuLim (kNm)	102	102	102	102	102	102
R	0.554	0.917	0.806	2.738	0.609	2.118
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.201	0.2	0.672	0.2	0.497
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	170.63	170.2	571.85	170.2	423.22
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural

				reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	64.32	33.9	56.73
Tu (kNm)	0.17	0.17	0
V_{Tu} (kN)	1.19	1.19	0
Vut (kN)	65.51	35.09	56.73
Tv (N/sqmm)	0.77	0.41	0.67
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	17.66	0	8.88
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	209.972	114.782	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G6**	
Beam No	:	**B19**	
Analysis Reference (Member)	:	87 (8.077 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	13	13	13	15
Critical L/C - RCDC	16	9	9	9	9	11
Mu (kNm)	18.03	35.19	33.26	87.59	18.64	64.52
Tu (kNm)	0	0.65	0.65	0.65	0.65	0.28
M_{Tu} (kNm)	0	1.1	1.1	1.1	1.1	0.47
Mud (kNm)	18.03	36.29	34.36	88.69	19.74	65
MuLim (kNm)	102	102	102	102	102	102
R	0.573	1.153	1.091	2.817	0.627	2.064
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.255	0.241	0.696	0.2	0.483
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	0.709	0.709	0.709
AstCalc (sqmm)	170.2	217.23	205	592.02	170.2	411.01
AstPrv (sqmm)	603.18	603.18	603.18	603.18	603.18	603.18
Reinforcement Provided	3-T16	3-T16	3-T16	3-T16	3-T16	3-T16

Note: Calculation of Ast

AstCalc	=	Max {B',C} x B X d (for Mu > 0)		
AstCalc	=	Bn x B X d (for Mu = 0)		
Where,				
A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	0.709	0.709	0.709
Vu (kN)	66.51	36.55	56.07
Tu (kNm)	0.65	0.65	0.28
V_{Tu} (kN)	4.53	4.53	1.95
Vut (kN)	71.05	41.09	58.02
Tv (N/sqmm)	0.83	0.48	0.68
Tc (N/sqmm)	0.56	0.56	0.56
Vc (kN)	47.85	47.85	47.85
Vus=Vut-Vc (kN)	23.2	0	10.17
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	241.273	147.514	189.72
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B20**	
Analysis Reference (Member)	:	127 (8.077 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	14	12	12	12	5	5
Critical L/C - RCDC	10	8	8	8	1	1
Mu (kNm)	3.36	-	-	12.98	69.74	97.83
Tu (kNm)	4.51	5.1	5.1	5.1	0	0
M_{Tu} (kNm)	7.61	8.61	8.61	8.61	0	0
Mud (kNm)	10.97	8.61	8.61	21.59	69.74	97.83
MuLim (kNm)	102	102	102	102	102	102
R	0.348	0.273	0.273	0.686	2.215	3.107
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.523	0.786
Pcclc (%) (C)	0.028	0.058	0.058	0	0	0

PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.46	668.74
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	44.84	51.08	53.09
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	44.84	51.09	53.09
Tv (N/sqmm)	0.53	0.6	0.62
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B21**	
Analysis Reference (Member)	:	112 (8.077 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	43.26	33.16	51.35	84.12	31.75	62.49
Tu (kNm)	0.11	0.17	0.15	0.12	0.17	0
M_{Tu} (kNm)	0.18	0.28	0.26	0.21	0.28	0
Mud (kNm)	43.44	33.45	51.61	84.32	32.03	62.49
MuLim (kNm)	102	102	102	102	102	102
R	1.38	1.062	1.639	2.678	1.017	1.985
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2

Ptclc (%) (B')	0.31	0.234	0.373	0.654	0.224	0.462
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	263.42	199.22	317.7	556.86	190.34	393.11
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	56.38	39.08	46.09
Tu (kNm)	0.12	0.12	0
V_{Tu} (kN)	0.86	0.86	0
Vut (kN)	57.24	39.95	46.09
Tv (N/sqmm)	0.67	0.47	0.54
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0.55	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	185.004	130.263	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B22**	
Analysis Reference (Member)	:	100 (8.077 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	20	13	18	8	13	15
Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	29.56	29.68	40.53	83.3	28.39	54.44

Tu (kNm)	0	0.17	0.13	0.18	0.17	0
M_{Tu} (kNm)	0	0.28	0.21	0.3	0.28	0
Mud (kNm)	29.56	29.96	40.74	83.6	28.67	54.45
MuLim (kNm)	102	102	102	102	102	102
R	0.939	0.952	1.294	2.655	0.911	1.729
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.206	0.208	0.289	0.648	0.2	0.396
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	174.89	177.42	245.79	551.11	170.2	337.04
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	53.05	33.8	42.36
Tu (kNm)	0.18	0.18	0
V_{Tu} (kN)	1.24	1.24	0
Vut (kN)	54.29	35.05	42.36
Tv (N/sqmm)	0.64	0.41	0.5
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	177.344	116.431	-
Asv Torsion Prv (sqmm)	365.6	365.6	-

Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G7**	
Beam No	:	**B23**	
Analysis Reference (Member)	:	88 (8.077 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	8	13	15

Critical L/C - RCDC	16	9	14	4	9	11
Mu (kNm)	26.69	36.55	47.59	80.06	25.34	52.06
Tu (kNm)	0	0.18	0.14	0.16	0.18	0
M_{Tu} (kNm)	0	0.3	0.23	0.27	0.3	0
Mud (kNm)	26.69	36.85	47.82	80.33	25.64	52.06
MuLim (kNm)	102	102	102	102	102	102
R	0.848	1.17	1.519	2.551	0.814	1.653
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.259	0.343	0.617	0.2	0.377
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	220.79	292.29	525.44	170.2	320.76
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	8	10
Critical L/C - RCDC	4	4	6
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	53.79	34.73	41.14
Tu (kNm)	0.16	0.16	0
V_{Tu} (kN)	1.09	1.09	0
Vut (kN)	54.89	35.83	41.14
Tv (N/sqmm)	0.64	0.42	0.48
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8

Asv Torsion (sqmm)	178.573	118.239	-
Asv Torsion Prv (sqmm)	365.6	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G8**	
Beam No	:	**B24**	
Analysis Reference (Member)	:	128 (8.077 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Cantilever Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design						
	Beam Bottom			**Beam Top**		

	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	7	12	12	7	5	5
Critical L/C - RCDC	3	8	8	3	1	1
Mu (kNm)	8.82	-	-	10.11	57.74	82.89
Tu (kNm)	6.6	6.64	6.64	6.6	3.58	3.58
M_{Tu} (kNm)	11.14	11.21	11.21	11.14	6.04	6.04
Mud (kNm)	19.96	11.21	11.21	21.25	63.78	88.93
MuLim (kNm)	102	102	102	102	102	102
R	0.634	0.356	0.356	0.675	2.026	2.824
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.473	0.698
Pcclc (%) (C)	0.015	0.076	0.076	0.007	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	170.2	170.2	170.2	402.32	593.96
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	7	7	5
Critical L/C - RCDC	3	3	1
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	37.1	40.59	47.64
Tu (kNm)	6.6	6.6	3.58
V_{Tu} (kN)	45.92	45.92	24.89
Vut (kN)	83.02	86.51	72.53
Tv (N/sqmm)	0.98	1.02	0.85
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	26.33	29.82	15.84

Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	465.69	476.728	339.535
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G8**
Beam No : **B25**
Analysis Reference (Member) : 113 (8.077 m)
Beam Length : 4496 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456
Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm

Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	35.01	32.88	39.99	96.13	30.39	78.35
Tu (kNm)	0.62	0.4	0.15	0.4	0.4	0.88
M_{Tu} (kNm)	1.05	0.68	0.25	0.68	0.68	1.48
Mud (kNm)	36.06	33.56	40.24	96.82	31.07	79.84
MuLim (kNm)	102	102	102	102	102	102
R	1.145	1.066	1.278	3.075	0.987	2.536
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.254	0.235	0.285	0.776	0.217	0.613
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	215.77	199.93	242.54	660.03	184.33	521.63
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	**Mid**	**Right**
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	68.63	41.64	62.3
Tu (kNm)	0.4	0.4	0.88

V_{Tu} (kN)	2.81	2.81	6.12
Vut (kN)	71.44	44.46	68.42
Tv (N/sqmm)	0.84	0.52	0.8
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	14.75	0	11.73
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	238.578	153.159	243.587
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group : **G8**
Beam No : **B26**
Analysis Reference (Member) : 101 (8.077 m)
Beam Length : 5067 mm
Breadth (B) : 230 mm
Depth (D) : 430 mm
Effective Depth (d) : 370 mm
Design Code : IS 456

Beam Type : Regular Beam
Grade Of Concrete (Fck) : M25 N/sqmm
Grade Of Steel (Main) : Fe550 N/sqmm
Grade Of Steel (Shear) : Fe550 N/sqmm
Top/Bottom Clear Cover (Cmin) : 30 mm
Side Clear Cover : 30 mm
Es : 2x10^5 N/sqmm
Ptnominal (%) (Bn) : 0.2 %

Flexure Design						
	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	18	13	13	15
Critical L/C - RCDC	16	9	14	9	9	11
Mu (kNm)	24.92	35.22	34.23	98.42	24.99	76.83
Tu (kNm)	0.15	0.37	0.31	0.37	0.37	0
M_{Tu} (kNm)	0.25	0.62	0.53	0.62	0.62	0
Mud (kNm)	25.16	35.84	34.76	99.04	25.61	76.83
MuLim (kNm)	102	102	102	102	102	102
R	0.799	1.138	1.104	3.146	0.813	2.44
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.252	0.244	0.798	0.2	0.586
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	214.35	207.51	679.32	170.2	498.51
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	13	15
Critical L/C - RCDC	4	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	69.65	38.98	60.86
Tu (kNm)	0.34	0.37	0
V_{Tu} (kN)	2.4	2.56	0
Vut (kN)	72.06	41.53	60.86
Tv (N/sqmm)	0.85	0.49	0.72
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	15.37	0	4.17
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	238.703	142.75	-
Asv Torsion Prv (sqmm)	718.14	365.6	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1	= 180	mm
Spc4=(X1+Y1)/4	= 140	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group : **G8**
Beam No : **B27**

Analysis Reference (Member)	:	89 (8.077 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	20	13	13	13	13	15
Critical L/C - RCDC	16	9	9	9	9	11
Mu (kNm)	25.95	41.6	41.58	98.4	23.7	78.11
Tu (kNm)	0.73	1.12	1.12	1.12	1.12	1.2
M_{Tu} (kNm)	1.24	1.89	1.89	1.89	1.89	2.02
Mud (kNm)	27.19	43.49	43.47	100.29	25.59	80.13
MuLim (kNm)	102	102	102	102	102	102
R	0.863	1.381	1.381	3.185	0.813	2.545
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.31	0.31	0.811	0.2	0.616
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.709	0.709	0.709	1.107	1.107	1.107
AstCalc (sqmm)	170.2	263.7	263.6	690.22	170.2	523.93
AstPrv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement Provided	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural

				reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	13	13	15
Critical L/C - RCDC	9	9	11
PtPrv (%)	1.107	1.107	1.107
Vu (kN)	71.11	41.15	61.3
Tu (kNm)	1.12	1.12	1.2
V_{Tu} (kN)	7.78	7.78	8.34
Vut (kN)	78.9	48.94	69.63
Tv (N/sqmm)	0.93	0.58	0.82
Tc (N/sqmm)	0.67	0.67	0.67
Vc (kN)	56.69	56.69	56.69
Vus=Vut-Vc (kN)	22.21	0	12.94
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	284.133	189.303	257.252
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm

Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)

Spc3 = X1 = 180 mm

Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm

Beam Depth = 430 mm

Web Depth = 430 <= 750 mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B28**	
Analysis Reference (Member)	:	165 (8.077 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	7	7	7	8	8	8
Critical L/C - RCDC	3	3	3	4	4	4
Mu (kNm)	-	-	-	10.34	24.15	34.77
Tu (kNm)	7.94	7.94	7.94	7.61	7.61	7.61
M_{Tu} (kNm)	13.41	13.41	13.41	12.85	12.85	12.85
Mud (kNm)	13.41	13.41	13.41	23.18	36.99	47.61
MuLim (kNm)	102	102	102	102	102	102
R	0.426	0.426	0.426	0.736	1.175	1.512
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.261	0.342
Pcclc (%) (C)	0.091	0.091	0.091	0.017	0	0
PtPrv (%)	0.709	0.709	0.709	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	221.72	290.92
AstPrv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement Provided	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	8	8	8
Critical L/C - RCDC	4	4	4
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	9.64	16.26	19.32
Tu (kNm)	7.61	7.61	7.61
V_{Tu} (kN)	52.94	52.94	52.94
Vut (kN)	62.58	69.19	72.26
Tv (N/sqmm)	0.74	0.81	0.85
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	24.7	31.31	34.38
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	416.796	437.492	447.086
Asv Torsion Prv (sqmm)	718.14	718.14	718.14
Asv Reqd (sqmm/m)	254.81	254.81	257.32
SvCalc (mm)	140	140	140
SvPrv (mm)	140	140	140
Asv Total Prv (sqmm)	718.14	718.14	718.14

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B29**	
Analysis Reference (Member)	:	166 (8.077 m)	
Beam Length	:	4496	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Critical L/C - Analysis	20	15	15	8	13	15
Critical L/C - RCDC	16	11	11	4	9	11
Mu (kNm)	7.52	7.73	-	38.58	15.98	17.31
Tu (kNm)	1.22	1.58	1.58	0.28	0.2	1.58
M_{Tu} (kNm)	2.06	2.66	2.66	0.46	0.34	2.66
Mud (kNm)	9.59	10.39	2.66	39.05	16.32	19.97
MuLim (kNm)	102	102	102	102	102	102
R	0.304	0.33	0.085	1.24	0.518	0.634
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.276	0.2	0.2
Pcclc (%) (C)	0	0	0.018	0	0	0

PtPrv (%)	0.709	0.709	0.709	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	234.87	170.2	170.2
AstPrv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement Provided	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	8	15
Critical L/C - RCDC	1	4	11
PtPrv (%)	0.399	0.399	0.399
Vu (kN)	26.19	12.87	19.39
Tu (kNm)	1.03	0.28	1.58
V_{Tu} (kN)	7.18	1.91	10.97
Vut (kN)	33.37	14.78	30.36
Tv (N/sqmm)	0.39	0.17	0.36
Tc (N/sqmm)	0.45	0.45	0.45
Vc (kN)	37.88	37.88	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	134.398	54.248	140.822
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B30**	
Analysis Reference (Member)	:	167 (8.077 m)	
Beam Length	:	5067	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	10	5	13	5	13	5
Critical L/C - RCDC	6	1	9	1	9	1
Mu (kNm)	5.25	13.37	6.27	21.18	-	19.57
Tu (kNm)	0.15	0.51	0.67	0.51	0.67	0.51
M_{Tu} (kNm)	0.25	0.86	1.14	0.86	1.14	0.86
Mud (kNm)	5.5	14.23	7.41	22.04	1.14	20.43
MuLim (kNm)	102	102	102	102	102	102
R	0.175	0.452	0.235	0.7	0.036	0.649
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2

Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.008	0
PtPrv (%)	0.709	0.709	0.709	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement Provided	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.399	0.709	0.399
Vu (kN)	26.09	9.26	25.46
Tu (kNm)	0.51	0.51	0.51
V_{Tu} (kN)	3.54	3.54	3.54
Vut (kN)	29.64	12.81	29
Tv (N/sqmm)	0.35	0.15	0.34
Tc (N/sqmm)	0.45	0.56	0.45
Vc (kN)	37.88	47.85	37.88
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	107.545	54.872	105.55
Asv Torsion Prv (sqmm)	365.6	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	275	275	275
SvPrv (mm)	275	275	275
Asv Total Prv (sqmm)	365.6	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 278	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 430	mm
Web Depth	= 430 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G9**	
Beam No	:	**B31**	
Analysis Reference (Member)	:	168 (8.077 m)	
Beam Length	:	4991	mm
Breadth (B)	:	230	mm
Depth (D)	:	430	mm
Effective Depth (d)	:	370	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	8	5	5	15
Critical L/C - RCDC	1	1	4	1	1	11
Mu (kNm)	6.94	20.25	17.3	22.04	-	12.12
Tu (kNm)	2.71	2.71	2.28	2.71	2.71	1.89
M_{Tu} (kNm)	4.58	4.58	3.84	4.58	4.58	3.19
Mud (kNm)	11.52	24.83	21.14	26.62	4.58	15.31
MuLim (kNm)	102	102	102	102	102	102
R	0.366	0.789	0.671	0.845	0.146	0.486

Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.031	0
PtPrv (%)	0.709	0.709	0.709	0.399	0.399	0.399
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
AstPrv (sqmm)	603.18	603.18	603.18	339.3	339.3	339.3
Reinforcement Provided	3-T16	3-T16	3-T16	3-T12	3-T12	3-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement
B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn = Ptnominal = Nominal % area of reinforcement
C = Pcclc = % area of compression reinforcement required
D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.399	0.709	0.399
Vu (kN)	28.97	12.41	21.77
Tu (kNm)	2.72	2.72	2.72
V_{Tu} (kN)	18.88	18.88	18.88
Vut (kN)	47.85	31.29	40.66
Tv (N/sqmm)	0.56	0.37	0.48
Tc (N/sqmm)	0.45	0.56	0.45
Vc (kN)	37.88	47.85	37.88
Vus=Vut-Vc (kN)	9.97	0	2.78
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	228.582	176.751	206.058
Asv Torsion Prv (sqmm)	718.14	365.6	365.6
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	140	275	275
SvPrv (mm)	140	275	275
Asv Total Prv (sqmm)	718.14	365.6	365.6

Maximum Spacing Criteria

Basic

Spc1 =0.75d = 278 mm
Spc2 = 300 mm

For Torsion

(X1 = 180, Y1 = 380)
Spc3 = X1 = 180 mm
Spc4=(X1+Y1)/4 = 140 mm

SFR Design

Beam Width = 230 mm
Beam Depth = 430 mm
Web Depth = 430 <= 750 mm
Side Face Reinforcement not required.